ORIGINAL ARTICLE

Trastuzumab Deruxtecan in Previously Treated HER2-Positive Gastric Cancer

K. Shitara, Y.-J. Bang, S. Iwasa, N. Sugimoto, M.-H. Ryu, D. Sakai, H.-C. Chung, H. Kawakami, H. Yabusaki, J. Lee, K. Saito, Y. Kawaguchi, T. Kamio, A. Kojima, M. Sugihara, and K. Yamaguchi, for the DESTINY-Gastric01 Investigators*

ABSTRACT

BACKGROUND

Trastuzumab deruxtecan (DS-8201) is an antibody-drug conjugate consisting of an anti-HER2 (human epidermal growth factor receptor 2) antibody, a cleavable tetrapeptide-based linker, and a cytotoxic topoisomerase I inhibitor. The drug may have efficacy in patients with HER2-positive advanced gastric cancer.

METHODS

In an open-label, randomized, phase 2 trial, we evaluated trastuzumab deruxtecan as compared with chemotherapy in patients with HER2-positive advanced gastric cancer. Patients with centrally confirmed HER2-positive gastric or gastroesophageal junction adenocarcinoma that had progressed while they were receiving at least two previous therapies, including trastuzumab, were randomly assigned in a 2:1 ratio to receive trastuzumab deruxtecan (6.4 mg per kilogram of body weight every 3 weeks) or physician's choice of chemotherapy. The primary end point was the objective response, according to independent central review. Secondary end points included overall survival, response duration, progression-free survival, confirmed response (response persisting ≥4 weeks), and safety.

RESULTS

Of 187 treated patients, 125 received trastuzumab deruxtecan and 62 chemotherapy (55 received irinotecan and 7 paclitaxel). An objective response was reported in 51% of the patients in the trastuzumab deruxtecan group, as compared with 14% of those in the physician's choice group (P<0.001). Overall survival was longer with trastuzumab deruxtecan than with chemotherapy (median, 12.5 vs. 8.4 months; hazard ratio for death, 0.59; 95% confidence interval, 0.39 to 0.88; P=0.01, which crossed the prespecified O'Brien–Fleming boundary [0.0202 on the basis of number of deaths]). The most common adverse events of grade 3 or higher were a decreased neutrophil count (in 51% of the trastuzumab deruxtecan group and 24% of the physician's choice group), anemia (38% and 23%, respectively), and decreased white-cell count (21% and 11%). A total of 12 patients had trastuzumab deruxtecan–related interstitial lung disease or pneumonitis (grade 1 or 2 in 9 patients and grade 3 or 4 in 3), as adjudicated by an independent committee. One drug-related death (due to pneumonia) was noted in the trastuzumab deruxtecan group; no drug-related deaths occurred in the physician's choice group.

CONCLUSIONS

Therapy with trastuzumab deruxtecan led to significant improvements in response and overall survival, as compared with standard therapies, among patients with HER2-positive gastric cancer. Myelosuppression and interstitial lung disease were the notable toxic effects. (Funded by Daiichi Sankyo; DESTINY-Gastric01 ClinicalTrials.gov number, NCT03329690.)

The authors' full names, academic degrees, and affiliations are listed in the Appendix. Address reprint requests to Dr. Shitara at the National Cancer Center Hospital East, 6-5-1, Kashiwanoha, Kashiwa, Chiba, 277-8577, Japan, or at kshitara@east.ncc.go.jp.

*A list of the DESTINY-Gastric01 Investigators is provided in the Supplementary Appendix, available at NEJM.org.

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An estimated 15 to 20% of advanced gastric and gastroesophageal junction cancers, which are especially prevalent in East Asia,[1] have overexpression or amplification of human epidermal growth factor receptor 2 (HER2).[2] Chemotherapy plus the anti-HER2 antibody trastuzumab is the recommended first-line therapy[3,4] on the basis of the randomized, multicenter, phase 3 ToGA (Trastuzumab for Gastric Cancer) trial, in which overall survival was significantly longer with chemotherapy plus trastuzumab than with chemotherapy alone (median, 13.8 months vs. 11.1 months).[5] Regardless of HER2 expression, paclitaxel plus ramucirumab (an anti–VEGFR-2 [vascular endothelial growth factor receptor 2] antibody) is recommended as second-line therapy[3] and is associated with a median overall survival of 9.6 months (vs. 7.4 months with paclitaxel alone), with somewhat prolonged progression-free survival (median, 4.4 months vs. 2.9 months) and higher response (28% vs. 16% of the patients).[6] For later lines of treatment, irinotecan, taxanes, trifluridine–tipiracil, and immune checkpoint inhibitors are additional options.[3,4] As compared with placebo, nivolumab[7] minimally prolonged overall survival (4.1 months vs. 5.3 months), as did trifluridine–tipiracil[8] (3.6 months vs. 5.7 months). Furthermore, several HER2-targeting agents did not prolong overall survival in patients with gastric cancer.[9-12]

Trastuzumab deruxtecan (DS-8201; known as [fam-]trastuzumab deruxtecan-nxki in the United States), an antibody-drug conjugate consisting of a humanized, monoclonal, anti-HER2 antibody bound to a cytotoxic topoisomerase I inhibitor (drug payload) by means of a cleavable, tetrapeptide-based linker, has been approved in the United States and Japan for the treatment of patients with metastatic HER2-positive breast cancer.[13] The linker is stable in plasma[13,14] but after internalization is cleaved by lysosomal enzymes such as cathepsins,[15] which are overexpressed in cancer cells.[16-18] With a drug-to-antibody ratio of approximately 8 and the ability of the released payload to diffuse across membranes and enter neighboring tumor cells, trastuzumab deruxtecan does not necessarily require high-level HER2 expression.[13,19] This may be advantageous in the treatment of gastric cancer, which often has heterogeneous HER2 expression.[2,20,21]

A recent phase 1 study (DS8201-A-J101; Clinical Trials.gov number, NCT02564900) assessed treatment with trastuzumab deruxtecan at a dose of 5.4 mg or 6.4 mg per kilogram of body weight in 44 patients with HER2-positive gastric or gastroesophageal junction cancer who had been previously treated with trastuzumab.[22] Positivity for HER2 was defined as a score of 3+ (positive) on immunohistochemical analysis or a score of 2+ (borderline) with positive results on in situ hybridization (such results are designated as high-level HER2 expression in this article).[22] The objective response according to investigator assessment was 43.2%, the median duration of response was 7.0 months, and the median progression-free survival was 5.6 months. Among patients who had previously received irinotecan, a different topoisomerase I inhibitor, 41.7% had a response, which suggests minimal cross-resistance with the trastuzumab deruxtecan payload. Here, we share the results of an open-label, multicenter, phase 2 trial of trastuzumab deruxtecan in patients with HER2-expressing advanced gastric cancer or gastroesophageal junction adenocarcinoma that had progressed after the receipt of at least two previous therapies.

METHODS

TRIAL DESIGN

We conducted a randomized, phase 2 trial involving patients with HER2-expressing, locally advanced or metastatic gastric or gastroesophageal junction cancer that had progressed after the patient had received at least two previous regimens, which included a fluoropyrimidine, a platinum agent, and trastuzumab (or approved biosimilar agent) (Fig. S1 in the Supplementary Appendix, available with the full text of this article at NEJM.org). HER2 levels were documented as high (score of 3+ on immunohistochemical analysis or score of 2+ with positive results on in situ hybridization) or low (score of 2+ with negative results on in situ hybridization or a score of 1+ [negative]).

The primary cohort consisted of patients with high-level HER2-positive disease and is the focus of this analysis. Patients with low-level HER2 disease were included in two separate exploratory cohorts; results are not reported here. Eligible patients were at least 20 years of age and had an Eastern Cooperative Oncology Group (ECOG) performance-status score of 0 or 1 (on a scale from 0 to 5, with higher scores indicating greater dis-

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ability). Patients were excluded if they had interstitial lung disease or pneumonitis or were suspected to have interstitial lung disease or pneumonitis that could not be ruled out on imaging at screening or if they had a history of noninfectious interstitial lung disease or pneumonitis that had been treated with glucocorticoids. The complete inclusion and exclusion criteria are listed in the Supplementary Appendix and are also available in the protocol, available at NEJM.org.

Patients were randomly assigned in a 2:1 ratio to receive either trastuzumab deruxtecan or the treating physician's choice of irinotecan or paclitaxel. Randomization was stratified according to country (Japan or South Korea), ECOG performance-status score (0 or 1), and HER2 status according to the guidelines from the American Society of Clinical Oncology–College of American Pathologists guidelines (score of 3+ on immunohistochemical analysis or score of 2+ with positive results on in situ hybridization).[23]

Patients in the trastuzumab deruxtecan group received the drug at a dose of 6.4 mg per kilogram of body weight, administered by intravenous infusion every 3 weeks, on the basis of previous analyses of pharmacokinetics, efficacy, and safety.[22] The standard dose that is used in patients with breast cancer is 5.4 mg per kilogram. Patients in the physician's choice group received either irinotecan monotherapy, at a dose of 150 mg per square meter of body-surface area administered every 2 weeks, or paclitaxel monotherapy, at a dose of 80 mg per square meter administered on days 1, 8, and 15 every 4 weeks. Treatment continued until disease progression, withdrawal of patient consent, or the occurrence of unacceptable adverse events.

TRIAL OVERSIGHT

The trial was sponsored and designed by Daiichi Sankyo and was approved by the institutional review board at each participating site. All the patients provided written informed consent. Data were analyzed and interpreted by the sponsor and authors. All the authors reviewed the manuscript and vouch for the accuracy of the data and analysis and for the adherence of the trial to the protocol. A professional writer participated in the preparation of the manuscript and was paid by the sponsor. In March 2019, AstraZeneca entered into a collaboration agreement with Daiichi Sankyo for trastuzumab deruxtecan. AstraZeneca was not involved in data collection but was involved in data interpretation as well as review of the manuscript and approval of its submission for publication.

END POINTS

The primary end point was the objective response (complete or partial) as determined according to independent central review with the use of the Response Evaluation Criteria in Solid Tumors, version 1.1. Overall survival was a key secondary end point to be statistically evaluated hierarchically if the primary end point was significant. Additional secondary end points were the duration of response, progression-free survival, confirmed disease control (confirmed objective response plus stable disease), confirmed objective response (objective response confirmed on a follow-up scan ≥4 weeks after the initial response as designated by an independent committee), and safety. Exploratory end points included the time to response and the best percentage change in the sum of the diameters of measurable tumors.

SAFETY

Adverse events were graded with the use of the National Cancer Institute Common Terminology Criteria for Adverse Events, version 4.03, and coded according to the *Medical Dictionary for Regulatory Activities*, version 20.1. Cases of potential interstitial lung disease or pneumonitis were evaluated by an independent adjudication committee that was separate from the response-assessment committee. The safety analysis set included all the patients who received at least one dose of the trial drug.

STATISTICAL ANALYSIS

We calculated that a sample of 180 patients (120 in the trastuzumab deruxtecan group and 60 in the physician's choice group) would provide the trial with 92.9% power to detect a difference in response between the two groups (assuming percentages of patients with response of 40% in the trastuzumab deruxtecan group and 15% in the physician's choice group) at a two-sided significance level of 0.05. Approximately 133 deaths would need to occur in order for the trial to have 80% power to detect a hazard ratio of 0.61 (64% improvement in median overall survival, from 5.5 to 9.0 months in the physician's choice group vs.

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the trastuzumab deruxtecan group) at a two-sided significance level of 0.05.

Response was compared between the treatment groups by means of Cochran–Mantel–Haenszel tests stratified according to region in the population of patients who had a response that could be evaluated (all the patients who underwent randomization, received at least one dose of trial drug, and had measurable tumors at baseline according to the independent review committee). Fisher's exact test between treatment groups was performed as a sensitivity analysis (unstratified). Overall survival in each treatment group was compared by stratified log-rank tests with region as a stratification factor in the full analysis set (all the patients who underwent randomization and received at least one dose of the trial drug).

The familywise type I error was controlled at 0.05 for the primary and secondary efficacy end points of objective response and overall survival by a fixed-sequence testing procedure in which, if the hypothesis test for objective response was significant at a two-sided alpha level of 0.05, overall survival was to be tested at the same level. The objective response analysis and interim analyses of overall survival were conducted after all the patients had undergone tumor assessment at approximately 24 weeks or had discontinued the trial; the final overall survival analysis was to be performed after approximately 133 deaths had occurred. The significance levels for the overall survival analyses were determined with the use of a Lan–DeMets alpha-spending function with an O'Brien–Fleming boundary. Hazard ratios and corresponding 95% confidence intervals were estimated with the use of Cox proportional-hazards regression models, stratified according to region, for the analyses of overall survival and progression-free survival; unadjusted hazard ratios with 95% confidence intervals (unstratified) are also reported.

The percentage of patients with confirmed disease control (confirmed objective response plus stable disease) is also reported. The distribution of time-to-event end points for overall survival, progression-free survival, and response duration was estimated by the Kaplan–Meier method. The median with a two-sided 95% confidence interval for these measures was calculated by the Brookmeyer and Crowley method.[24]

RESULTS

PATIENTS

From November 2017 through May 2019, a total of 188 patients in the primary cohort underwent randomization at 48 sites in Japan and 18 sites in South Korea. A total of 187 patients received treatment; 125 patients were treated with trastuzumab deruxtecan and 62 with the physician's choice of therapy (55 patients received irinotecan, and 7 received paclitaxel) (Fig. S2). One patient in the trastuzumab deruxtecan group did not receive treatment because the patient was ineligible owing to an abnormal echocardiogram before the initiation of treatment. The characteristics of the patients were balanced across the two groups (Table 1).

Patients had received a median of two previous systemic therapies (range, two to nine) for advanced or metastatic disease, and 31 patients (17%) had received at least four previous therapies; 72% of the patients had previously received ramucirumab, and 86% had received taxanes. The median time since the last administration of trastuzumab was 5.9 months (range, 0.7 to 40.7) among patients in the trastuzumab deruxtecan group and 6.5 months (range, 0.7 to 23.0) among those in the physician's choice group.

At the data cutoff (November 8, 2019), a total of 22% of the patients in the trastuzumab deruxtecan group and 5% of those in the physician's choice group were continuing to receive treatment. The median duration of treatment was 4.6 months (range, 0.7 to 22.3) in the trastuzumab deruxtecan group and 2.8 months (range, 0.5 to 13.1) in the physician's choice group.

EFFICACY

Treatment with trastuzumab deruxtecan resulted in a significantly higher percentage of patients with an objective response (according to independent central review) than did physician's choice of chemotherapy (51% vs. 14%, $P<0.001$) (Table 2). A confirmed objective response (i.e., an objective response lasting ≥4 weeks, although all responses were determined by independent central review) occurred in 51 patients (43%) in the trastuzumab deruxtecan group, as compared with 7 (12%) in the physician's choice group. In this analysis, 10 of 119 patients in the trastuzumab deruxtecan group had a confirmed complete response, as

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Table 1. Demographic and Clinical Characteristics of the Patients at Baseline.*

Characteristic	Trastuzumab Deruxtecan (N=125)	Physician's Choice of Chemotherapy (N=62)
Median age (range) — yr†	65 (34–82)	66 (28–82)
Female sex — no. (%)	30 (24)	15 (24)
Region — no. (%)		
Japan	99 (79)	50 (81)
South Korea	26 (21)	12 (19)
ECOG performance-status score — no. (%)‡		
0	62 (50)	30 (48)
1	63 (50)	32 (52)
Histologic subtype — no. (%)		
Intestinal	89 (71)	38 (61)
Diffuse	28 (22)	18 (29)
Other	8 (6)	6 (10)
HER2 expression — no. (%)§		
IHC 3+	96 (77)	47 (76)
IHC 2+ or ISH-positive	29 (23)	15 (24)
Primary site — no. (%)		
Stomach	108 (86)	55 (89)
Gastroesophageal junction	17 (14)	7 (11)
Sum of diameters of measurable tumors — no. (%)¶		
<5 cm	63 (50)	26 (42)
≥5 cm to <10 cm	34 (27)	22 (35)
≥10 cm	22 (18)	8 (13)
Missing data	6 (5)	6 (10)
No. of previous systemic therapies for advanced or metastatic disease — no. (%)‖		
2	66 (53)	38 (61)
3	34 (27)	18 (29)
≥4	25 (20)	6 (10)
Previous treatment — no. (%)		
Therapy containing trastuzumab	125 (100)	62 (100)
Therapy containing taxane	105 (84)	55 (89)
Therapy containing ramucirumab	94 (75)	41 (66)
Irinotecan or other topoisomerase I inhibitor	8 (6)	5 (8)
Immune checkpoint inhibitor	44 (35)	17 (27)

* Shown are data for all treated patients. In the physician's choice group, 55 patients received irinotecan and 7 received paclitaxel. Percentages may not total 100 because of rounding.

† Shown is the median age at the time that informed consent was provided.

‡ Performance-status scores on the Eastern Cooperative Oncology Group (ECOG) scale range from 0 to 5, with higher scores indicating greater disability.

§ HER2 (human epidermal growth factor receptor 2) expression was centrally confirmed prospectively by analysis of the most recent archival tissue according to the guidelines from the American Society of Clinical Oncology–College of American Pathologists.[23] According to these guidelines, HER2 positivity was defined as a HER2 immunohistochemical (IHC) analysis of a score of 3+ (positive) or a score of 2+ (borderline) and positive results on in situ hybridization (ISH).

¶ Data were based on independent central review at baseline (the last nonmissing value obtained before the first dose of trial drug). Patients with missing data had measurable disease according to the investigator's assessment but had no measurable lesions according to independent central review.

‖ The previous therapies were intended to treat locally advanced or metastatic disease or were intended as neoadjuvant or adjuvant therapy if progressive disease occurred within 6 months since the end of the therapy.

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Table 2. Summary of Efficacy.*

Variable	Trastuzumab Deruxtecan (N=119)	Physician's Choice of Chemotherapy (N=56)
Objective response†		
No. of patients	61	8
Percent of patients (95% CI)	51 (42–61)	14 (6–26)
Best response — no. (%)		
Complete response	11 (9)	0
Partial response	50 (42)	8 (14)
Stable disease	42 (35)	27 (48)
Progressive disease	14 (12)	17 (30)
Could not be evaluated	2 (2)	4 (7)
Confirmed objective response‡		
No. of patients	51	7
Percent of patients (95% CI)	43 (34–52)	12 (5–24)
Confirmed best response — no. (%)		
Complete response	10 (8)	0
Partial response	41 (34)	7 (12)
Stable disease	51 (43)	28 (50)
Progressive disease	14 (12)	17 (30)
Could not be evaluated	3 (3)	4 (7)
Confirmed disease control§		
No. of patients	102	35
Percent of patients (95% CI)	86 (78–91)	62 (49–75)

* The analyses included all the patients who underwent randomization, received at least one dose of trial drug, and had measurable tumors as assessed on independent central review at baseline. In the physician's choice group, 51 patients received irinotecan and 5 received paclitaxel. Percentages may not total 100 because of rounding. CI denotes confidence interval.

† An objective response was defined as a complete or partial response. The comparison was evaluated by Cochran–Mantel–Haenszel tests stratified according to region (P<0.001). Fisher's exact test between treatment groups was performed as a sensitivity analysis (unstratified P<0.001).

‡ A confirmed objective response was defined as a complete or partial response that was confirmed on a follow-up scan performed at least 4 weeks after the initial complete or partial response was noted.

§ Confirmed disease control was defined as a confirmed objective response or stable disease.

compared with none of the 56 patients in the physician's choice group. Most patients (>80%) receiving trastuzumab deruxtecan had a reduction in tumor size, as compared with approximately half the patients receiving physician's choice of chemotherapy (Fig. 1 and Fig. S3).

The median duration of confirmed objective response was 11.3 months (95% confidence interval [CI], 5.6 to could not be estimated) in the trastuzumab deruxtecan group, as compared with 3.9 months (95% CI, 3.0 to 4.9) in the physician's choice group (Fig. S4). The median time to a response was similar in the two groups (1.5 months [95% CI, 1.4 to 1.7] in the trastuzumab deruxtecan group and 1.6 months [95% CI, 1.3 to 1.7] in the physician's choice group). Treatment with trastuzumab deruxtecan resulted in a higher percentage of patients with confirmed disease control (86%; 95% CI, 78 to 91) than did physician's choice of chemotherapy (62%; 95% CI, 49 to 75).

Overall survival was significantly longer in the trastuzumab deruxtecan group than in the physician's choice group (median, 12.5 months vs. 8.4 months; hazard ratio for death, 0.59; 95% CI, 0.39 to 0.88; P=0.01, which crossed the prespecified O'Brien–Fleming boundary [0.0202 on the basis of the number of deaths]) (Fig. 2A). Overall, 101 of 187 patients died: 62 of 125 patients (50%) in the trastuzumab deruxtecan group, and 39 of 62 (63%) in the physician's choice group. The estimated overall survival was 80% in the trastuzumab deruxtecan group and 66% in the physician's choice group at 6 months and was 52% and 29%, respectively, at 12 months. In the trastuzumab deruxtecan group, 60 patients (48%) received subsequent treatment after they discontinued the protocol treatment, as compared with 46 patients (74%) in the physician's choice group; inhibitors of programmed cell death protein 1 (PD-1) or its ligand (PD-L1) were used in 31% of the patients in the trastuzumab deruxtecan group and in 45% of those in the physician's choice group.

The median progression-free survival was 5.6 months (95% CI, 4.3 to 6.9) in the trastuzumab deruxtecan group and 3.5 months (95% CI, 2.0 to 4.3) in the physician's choice group (hazard ratio for progression or death, 0.47; 95% CI, 0.31 to 0.71) (Fig. 2B). The estimated progression-free survival was 43% in the trastuzumab deruxtecan group and 21% in the physician's choice group at 6 months and was 30% in the trastuzumab deruxtecan group at 12 months; no patients in the physician's choice group had progression-free survival at 12 months. A total of 73 patients (58%) in the trastuzumab deruxtecan group and 36 (58%) in the physician's choice group died or had disease progression according to independent central review.

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SUBGROUP ANALYSIS

In a prespecified subgroup analysis, the percentages of patients with an objective response were greater among patients receiving trastuzumab deruxtecan than among those receiving physician's choice of chemotherapy, and the findings were generally consistent within each treatment group across subgroups (Fig. S5). Among patients who received trastuzumab deruxtecan, the percentage of patients with an objective response was higher among those with a HER2 score of 3+ on immunohistochemical analysis than among those with a score of 2+ with positive results on in situ hybridization (53 of 91 patients [58%] vs. 8 of 28 [29%]). Overall survival also tended to favor trastuzumab deruxtecan over physician's choice of chemotherapy across most subgroups (Fig. S6).

SAFETY

All 125 patients in the trastuzumab deruxtecan group had at least one adverse event (of any grade) during the treatment phase of the trial, as compared with 61 of 62 patients (98%) in the physician's choice group (Table 3 and Tables S1 and S2). The most common adverse events of grade 3 or higher were a decreased neutrophil count (in 51% of the patients in the trastuzumab deruxtecan group and in 24% of those in the physician's choice group), anemia (in 38% and 23%, respectively), a decreased white-cell count (in 21% and 11%), and decreased appetite (in 17% and 13%). A total of 6 patients in the trastuzumab deruxtecan group had febrile neutropenia (all events of grade 3), as compared with 2 patients in the physician's choice group (one event each of grades 3 and 4).

A higher percentage of patients in the trastuzumab deruxtecan group than in the physician's choice group discontinued therapy (15% vs. 6%) or interrupted treatment owing to adverse events (62% vs. 37%), but the percentage of patients with dose reduction was similar in the two groups (32% and 34%, respectively). One death in the trastuzumab deruxtecan group was considered by the investigators to be related to therapy (due to pneumonia). This death occurred after the administration of cycle 6, and the patient did not have neutropenia at the time of the event. Details of the cause of pneumonia were not reported. No deaths in the physician's choice group were considered to be related to therapy.

A total of 12 patients (10%) in the trastuzumab deruxtecan group had drug-related interstitial lung disease or pneumonitis (median time

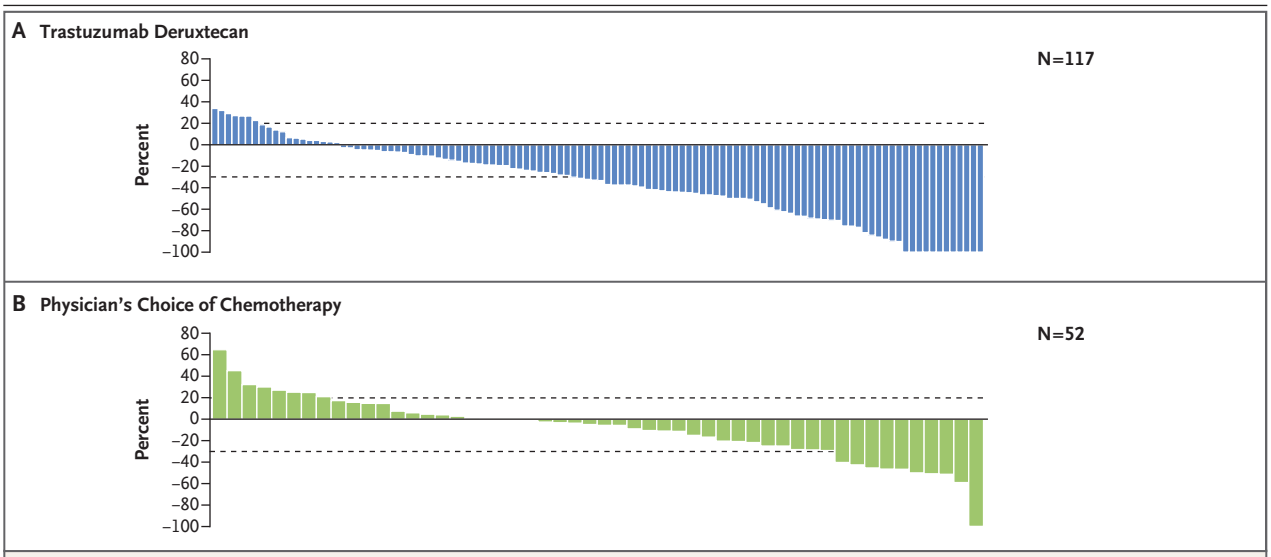

Figure 1. Best Percentage Change from Baseline in the Sum of Longest Diameters of Measurable Tumors.

The line at 20% indicates progressive disease, and the line at −30% indicates a partial response. The analyses included patients who had both baseline and postbaseline target-lesion assessments according to independent central review. Six patients (two in the trastuzumab deruxtecan group and four in the physician's choice group) were excluded from this analysis because they did not undergo postbaseline tumor assessment.

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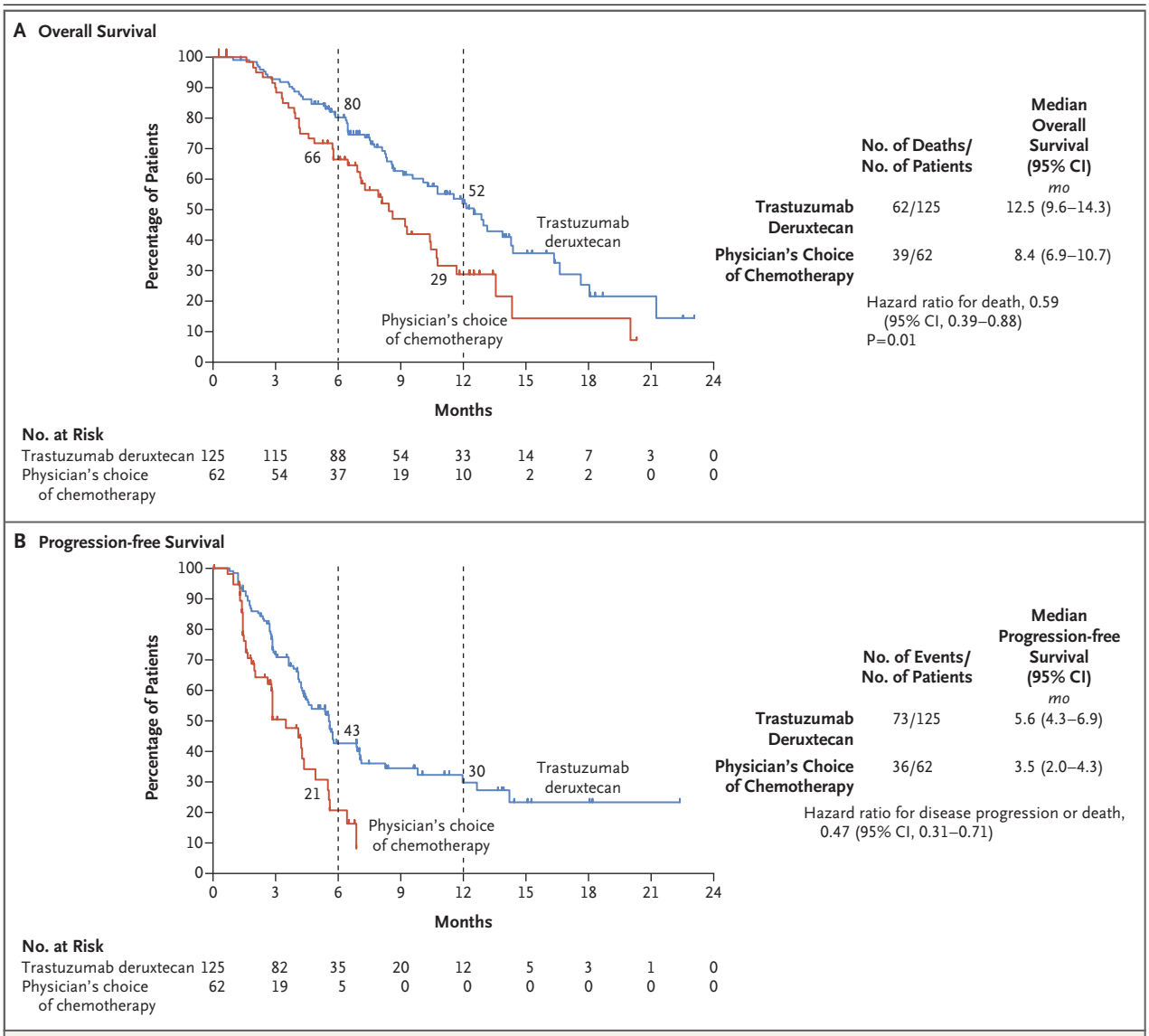

Figure 2. Overall Survival and Progression-free Survival.

In the analysis of overall survival (Panel A), 63 patients (50%) in the trastuzumab deruxtecan group and 23 (37%) in the physician's choice group had their data censored (tick marks). The two-sided P value of 0.01 crossed the O'Brien–Fleming boundary of significance (0.0202 on the basis of the number of deaths). The percentages of patients who survived are shown at 6 months and 12 months. In the analysis of progression-free survival (Panel B), 62 patients (50%) in the trastuzumab deruxtecan group had progressive disease, as assessed on independent central review, and 11 (9%) had death as the first event; the corresponding values in the physician's choice group were 34 patients (55%) and 2 (3%). A total of 52 patients (42%) in the trastuzumab deruxtecan group and 26 (42%) in the physician's choice group had their data censored (tick marks). Data were censored for the following reasons: no baseline tumor assessment (for 1 patient in the trastuzumab deruxtecan group and for 2 in the physician's choice group), no postbaseline tumor assessment (for 1 and 3, respectively), receipt of new anticancer therapy (for 6 and 11), and missing of two consecutive tumor assessments (for 3 and 1); the remaining patients had data censored without an event. The percentages of patients who were free from disease progression or death are shown at 6 months and 12 months. Both analyses were stratified according to region.

to first onset, 84.5 days; range, 36 to 638) as determined by an independent adjudication committee. Most events were of grade 1 or 2 (with three events of grade 1, six of grade 2, two of grade 3, one of grade 4, and no grade 5 events). At the time of this analysis, the majority of cases (eight) had resolved or were resolving (median duration, 57.0 days), three had not re-

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Table 3. Adverse Events Occurring in at Least 20% of the Patients Treated with Trastuzumab Deruxtecan.*

Preferred Term	Trastuzumab Deruxtecan (N=125)			Physician's Choice of Chemotherapy (N=62)		
	Any Grade	Grade 3	Grade 4	Any Grade	Grade 3	Grade 4
	number of patients (percent)					
Nausea	79 (63)	6 (5)	0	29 (47)	1 (2)	0
Neutrophil count decreased†	79 (63)	48 (38)	16 (13)	22 (35)	10 (16)	5 (8)
Decreased appetite	75 (60)	21 (17)	0	28 (45)	8 (13)	0
Anemia‡	72 (58)	47 (38)	0	19 (31)	13 (21)	1 (2)
Platelet count decreased§	49 (39)	12 (10)	2 (2)	4 (6)	1 (2)	1 (2)
White-cell count decreased¶	47 (38)	26 (21)	0	22 (35)	5 (8)	2 (3)
Malaise	43 (34)	1 (1)	0	10 (16)	0	0
Diarrhea	40 (32)	3 (2)	0	20 (32)	1 (2)	0
Vomiting	33 (26)	0	0	5 (8)	0	0
Constipation	30 (24)	0	0	14 (23)	0	0
Pyrexia	30 (24)	0	0	10 (16)	0	0
Alopecia	28 (22)	0	0	9 (15)	0	0
Fatigue	27 (22)	9 (7)	0	15 (24)	2 (3)	0
Lymphocyte count decreased‖	27 (22)	8 (6)	6 (5)	2 (3)	0	1 (2)

* No additional adverse events during the trial were observed in at least 20% of the patients receiving physician's choice of chemotherapy.
† This category includes the preferred terms neutrophil count decreased and neutropenia.
‡ This category includes the preferred terms hematocrit decreased, hemoglobin decreased, red-cell count decreased, and anemia.
§ This category includes the preferred terms platelet count decreased and thrombocytopenia.
¶ This category includes the preferred terms white-cell count decreased and leukopenia.
‖ This category includes the preferred terms lymphocyte count decreased and lymphopenia.

solved (one each of grades 1, 2, and 4), and the status of one was unknown (grade 2). No cases of interstitial lung disease occurred in the physician's choice group. A decrease in left ventricular ejection fraction or heart failure was not observed in either group.

DISCUSSION

This trial assessed the efficacy and safety of trastuzumab deruxtecan as compared with physician's choice of chemotherapy (irinotecan or paclitaxel) as third-line or later therapy in patients with HER2-positive gastric or gastroesophageal adenocarcinoma. The percentage of patients with an objective response was significantly higher in the trastuzumab deruxtecan group than in the physician's choice group (51% vs. 14%). A total of 10 patients in the trastuzumab deruxtecan group had a confirmed complete response, and overall survival was longer in the trastuzumab deruxtecan group than in the physician's choice group (median, 12.5 months vs. 8.4 months). The median duration of confirmed response was longer with trastuzumab deruxtecan than with physician's choice of chemotherapy (11.3 months vs. 3.9 months). The findings of this trial confirm those observed in a phase 1 trial of trastuzumab deruxtecan in patients with advanced HER2-positive gastric cancer (response according to investigator assessment, 43.2%; median progression-free survival, 5.6 months).[22]

The higher percentage of patients with an objective response and longer overall survival with trastuzumab deruxtecan than with chemotherapy were observed across subgroups. The efficacy of trastuzumab deruxtecan was greater than that of chemotherapy in the subgroup of patients with the highest level of HER2 expression (58% vs. 29%). However, the small number of patients in this subgroup makes definitive conclusions difficult.

Unlike in the treatment of metastatic breast cancer,[25,26] no HER2-directed agents other than trastuzumab have shown a significant benefit in patients with HER2-positive gastric cancer. Among

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patients with gastric cancer that had progressed, continued treatment with trastuzumab and paclitaxel did not prolong survival as compared with paclitaxel alone.[27] Pertuzumab combined with trastuzumab and chemotherapy in the context of first-line therapy did not further prolong overall or progression-free survival[9]; neither did lapatinib ditosylate combined with chemotherapy in a cohort that predominantly involved patients who had not received trastuzumab previously as first-line[11] or second-line[10] treatment, as compared with placebo or chemotherapy alone, respectively. In addition, chemotherapy and anti–PD-1 agents have resulted in limited benefits in these patients in the context of third-line therapy, with modest improvements in overall survival and generally low percentages of patients with a response (4% with trifluridine–tipiracil,[8] 11.2% with nivolumab,[7] and 11.6% with pembrolizumab[28]). Moreover, the anti–PD-L1 antibody avelumab did not prolong overall or progression-free survival, as compared with chemotherapy, nor did it result in a higher percentage of patients with a response.[29]

Acquired resistance to trastuzumab is common in patients who have previously been treated with this antibody. Multiple mechanisms have been suggested, including activating mutations in phosphoinositide 3-kinase (PI3K) that promote PI3K–AKT signaling as well as increased signaling through other growth factor receptors.[30,31] In addition, decreased HER2 expression after trastuzumab treatment has been observed.[27,30-32] Change in HER2 expression, combined with the heterogeneous HER2 expression noted in patients with gastric cancer,[2,20,21] may limit the efficacy of HER2-targeted treatments. Trastuzumab deruxtecan has shown activity in HER2-low breast cancer[33] and preclinically in cells with varying levels of HER2 expression.[13,19] The high drug-to-antibody ratio of trastuzumab deruxtecan and the membrane permeability of its payload may make trastuzumab deruxtecan less dependent on high HER2 expression, which may be particularly beneficial in the treatment of tumors with lower or heterogeneous HER2 expression.

In our trial, we assigned patients to cohorts on the basis of the HER2 status of their most recent archival tissue. We cannot preclude changes in HER2 status subsequent to the tissue sample we obtained. From the data obtained in this trial, lower levels of HER2 expression seem to be likely to result in lower percentages of patients with a response. Biomarker analyses of patients' tumor and liquid biopsy samples will be needed to provide information about the mechanism of action of trastuzumab deruxtecan and potential resistance mechanisms.

Trastuzumab emtansine, an antibody-drug conjugate with a different payload than trastuzumab deruxtecan (emtansine is a microtubule inhibitor), was tested in patients with gastric cancer that had progressed during or after treatment with trastuzumab; however, trastuzumab emtansine did not prolong median overall survival (7.9 months, vs. 8.6 months with taxanes) or lead to a higher percentage of patients with a response (20.6% vs. 19.6%).[12] Unlike with trastuzumab deruxtecan, the trastuzumab emtansine payload remains attached to its linker after the conjugated antibody is degraded within the host cell, and thus it is less membrane-permeable.[19,34] In addition, the topoisomerase I inhibitor payload of trastuzumab deruxtecan is approximately 10 times more potent than SN-38, the active metabolite of irinotecan.[13,35] Furthermore, the drug-to-antibody ratio of trastuzumab emtansine is lower than that of trastuzumab deruxtecan (approximately 3.4 vs. 8).[19,36] The more efficient linker and payload system of trastuzumab deruxtecan may contribute to the differing treatment outcomes.

In this trial, the notable adverse events occurring with trastuzumab deruxtecan were myelosuppression and interstitial lung disease, which were managed by appropriate dose reduction or interruption. Although most gastrointestinal events were of low grade, hematologic events were more frequently of grade 3 or higher with trastuzumab deruxtecan than with chemotherapy. These toxic effects were often addressed with dose modification. Although HER2-targeted therapies, including trastuzumab, have been associated with cardiotoxic effects,[37] this was not observed in our trial.

Interstitial lung disease is an important risk associated with irinotecan and HER2-targeted therapies, including trastuzumab, trastuzumab emtansine, and trastuzumab deruxtecan.[22,38,39] In the trastuzumab deruxtecan group, interstitial lung disease developed in 10% of the patients, as determined by an independent adjudicating committee. Most of these cases (9 of 12) were of low

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grade; 2 were grade 3, and 1 was grade 4. The signs and symptoms of interstitial lung disease were actively monitored and managed with dose modification or discontinuation, glucocorticoid therapy, and supportive care, in accordance with the trial protocol. Education about and close monitoring for signs and symptoms of interstitial lung disease (including fever, cough, or dyspnea) are recommended for early detection. If interstitial lung disease is suspected, useful evaluations include high-resolution computed tomography, consultation with a pulmonologist, and testing of pulmonary function and oxygen saturation, as well as other tests as needed.

Limitations of this trial include the limited ethnic diversity and relatively small sample size. However, in trials of other HER2-directed agents,[9,12] no significant difference in the treatment response was seen among patients from Asia as compared with those from other regions. Similarly, ethnic group–dependent differences in efficacy were not observed in a phase 2 trial of trastuzumab deruxtecan in patients with HER2-positive breast cancer.[38] Although the number of patients in this phase 2 trial was relatively small, the trial was adequately powered to detect prolonged overall survival under the prespecified statistical hypotheses. A single-group, phase 2 trial is under way in the United States and Europe to evaluate trastuzumab deruxtecan in the context of second-line therapy (after the receipt of first-line trastuzumab) (NCT04014075).

Overall, treatment with trastuzumab deruxtecan led to a significantly higher percentage of patients with an objective response and to longer overall survival than conventional chemotherapy among patients with HER2-positive, advanced gastric or gastroesophageal junction cancer. Antitumor activity was noted in patients who had had disease progression while they had been receiving regimens including trastuzumab. Noteworthy toxic effects included myelosuppression and interstitial lung disease, which were ameliorated by dose reductions and interruptions.

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APPENDIX

The authors' full names and academic degrees are as follows: Kohei Shitara, M.D., Yung-Jue Bang, M.D., Ph.D., Satoru Iwasa, M.D., Ph.D., Naotoshi Sugimoto, M.D., Ph.D., Min-Hee Ryu, M.D., Ph.D., Daisuke Sakai, M.D., Ph.D., Hyun-Cheol Chung, M.D., Ph.D., Hisato Kawakami, M.D., Ph.D., Hiroshi Yabusaki, M.D., Ph.D., Jeeyun Lee, M.D., Ph.D., Kaku Saito, M.Sc., Yoshinori Kawaguchi, M.Sc., Takahiro Kamio, M.D., Akihito Kojima, M.Sc., Masahiro Sugihara, Ph.D., and Kensei Yamaguchi, M.D.

The authors' affiliations are as follows: the National Cancer Center Hospital East, Kashiwa (K. Shitara), the National Cancer Center Hospital (S.I.), Daiichi Sankyo (T.K., A.K., M.S.), and the Cancer Institute Hospital of JFCR (K.Y.), Tokyo, the Osaka International Cancer Institute (N.S.), Osaka University Hospital (D.S.), and Kindai University Hospital (H.K.), Osaka, and Niigata Cancer Center Hospital, Niigata (H.Y.) — all in Japan; Seoul National University College of Medicine (Y.-J.B.), the Asan Medical Center, University of Ulsan College of Medicine (M.-H.R.), the Yonsei Cancer Center, Yonsei University College of Medicine (H.-C.C.), and the Samsung Medical Center, Sungkyunkwan University School of Medicine (J.L.) — all in Seoul, South Korea; and Daiichi Sankyo, Basking Ridge, NJ (K. Saito, Y.K.).

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